

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

22nd May 2013

Subject ST7 – General Insurance: Reserving & Capital Modelling

Time allowed: Three Hours (14.45* – 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.***
- 2. *Mark allocations are shown in brackets.***
- 3. *Attempt all questions, beginning your answer to each question on a separate sheet.***

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1) i)** What is an economic capital model and how it is different from standard stochastic models? (2)

A large international general insurer writes both personal lines and commercial lines business. The company has invested significant amount of human and financial capital to build an economic capital model, required to meet the new solvency requirements.

Historically, the key performance bonus measure for managers was based on the Gross Written Premium and Gross Underwriting Profit.

- ii)** Suggest and discuss the possible new measure that can be used using the economic capital model output that can have a direct impact on the share prices of the company. (3)

Given the success of new performance measure, managing director has asked you to suggest other key areas where the model can be used.

- iii)** Discuss in detail how economic capital model can be used to efficiently run the company apart from regulatory solvency requirements to improve the results. (6)

[11]

- Q. 2)** MASA is launching a space shuttle to Mars to investigate the existence of water and is looking for accidental damage insurance for next 25 years.

List the requirements for such risks to be insurable?

[5]

- Q. 3)** NEWX is a personal lines motor insurance company. Company has decided to underwrite large commercial property risks beginning 2014.

- i)** Discuss briefly the possible distribution channels relevant for large commercial property risks. (3)

- ii)** Explain the method of slip system used to write insurance contracts. (5)

Management has decided to set up a Data Group (both products) that will be responsible for quality of data that is being collected for pricing and reserving purposes.

- iii)** Suggest possible stakeholders that in your view should be part of this group and role each will play. (9)

[17]

- Q. 4)** A company writes personal lines motor and household insurance products and has recently completed the capital modeling exercise for the entire business. Capital requirement by risk type is as follows:

Risk Type	INR'Million
Reserving	200
Underwriting	300
Catastrophe	150
Credit	50
Market	50
Operational	250
Total before Diversification	1,000

- i) Discuss the possible suggestions on how to parameterize operational risk the company faces and issues that are associated with modeling the operational risk given the above capital assessment. (7)
- ii) Suggest possible measure to improve the capital requirement associated with operation risk. (2)

Company has signed a 5 year fixed price agreement with large affinity partner for its personal motor and household portfolio with high commission ratio (15% -20% higher than current commission paid to other tied agents).

- iii) Discuss how this agreement is likely to have an impact on capital required for various risk types? (4)
- [13]

- Q. 5)** Big Bang, a general insurance company, writes a large portfolio of group personal accident business. The group personal accident policy offers a fixed lump sum benefit of INR 200,000 in the event of accidental death of an insured member provided that the insured person suffers an accident within the policy period leading to death within three years from the date of the accident.

The following cumulative incurred claim development pattern for all group personal accident business:

Development factors (months)

12-24	24-36	36-48	48-60	60-72	72-84	84-96
1.60	1.27	1.15	1.03	1.01	1.00	1.00

The following two new product features were added to all policies renewing in 2014:

- A limitation period of one year on the date of intimation from the date of death.
- An additional benefit of INR 1000 per week from the date of the accident to the date of death.

- i) Comment on the suitability of using the same incurred development pattern across all group personal accident policies. (1)
 - ii) A large group policy is renewed every year on 1 January. Describe a possible approach to estimate the IBNR for the Accident Year 2014 for this account. (6)
 - iii) Describe the claim characteristics and the implications on the Asset – Liability matching strategy for the group personal accident business. (3)
- [10]**

Q. 6) A study of reporting delays and settlement delays for a class of business has shown that the:

- a) Proportion of claims reported in the n th year after the year of occurrence is given by: $1/(n+2)$ for $n = 0$ and 1 , 0 for $n > 2$.
- b) Proportion of claims settled in the n th year after the year of intimation is 50% for $n=0$ and 1 .

The following table gives the reported claim counts by accident year at December 31, 2012:

AY	Reported at Dec 31, 2012
2009	240
2010	200
2011	160
2012	120

For Accident Years 2009-2012,

- i) Calculate the number of claims settled in calendar year 2012 (3)
 - ii) Project the number of claims settled in calendar year 2013 (3)
- [6]**
- Q. 7)** A one-off construction project beginning on July 1 2013 is insured for a period of 3 years. Risk is linear with respect to time with exposure being zero at the beginning of the policy. The written premium is 300. Project the earned premiums by calendar year. (5)
- Q. 8)** Suggest various ways of communicating the extent of uncertainty in the reserves. (4)
- Q. 9)** A general insurance company has been writing large volumes of Private Car Motor Third Party Liability business for around 10 years. The book is protected by a combination of stable risk excess of loss treaty reinsurance and quota share treaty reinsurance programs. This treaty reinsurance is purchased annually with effect from 1 January. The risk excess of loss reinsurance policy is purchased on a losses occurring (LOD) basis and the quota share policy is purchased on a risks attaching (RAD) basis. The quota share operates before the risk excess of loss.

A court award on 1 January 2011 effectively increased the payout amounts towards accidental death of third party by roughly 20%. Discuss the considerations to be made in the following exercises:

- i) Gross of reinsurance reserving (8)
- ii) Reinsurance reserving (5)
- [13]
- Q. 10)** State examples of restrictions on underwriting that could be imposed by the regulator. [6]
- Q. 11) i)** State the two conditions under which payment under an Industry Loss Warranty (ILW) gets triggered. (2)
- ii) State the main considerations required in purchasing ILWs vis-à-vis catastrophe reinsurance. (3)
- [5]
- Q. 12)** Suggest ratios (i.e. give the numerator and denominator) and how they could be used by the management of a general insurer for monitoring the performance in terms of:
- claims experience
 - investment return
 - overall profitability of writing insurance
 - financial strength
 - rate of return on shareholders' capital
- [5]
